JACKFRUIT-LEVEL PROBLEM OS

SANPSHOTS OF THE OUTPUTS

1. Build success

```
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$ make clean
make
make -C /lib/modules/6.8.0-101-generic/build M=/home/abhijeet/OS-Jackfruit/boilerplate clean
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-101-generic'
  CLEAN   /home/abhijeet/OS-Jackfruit/boilerplate/Module.symvers
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-101-generic'
rm -f engine
make -C /lib/modules/6.8.0-101-generic/build M=/home/abhijeet/OS-Jackfruit/boilerplate modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-101-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.2) 12.3.0
  You are using:           gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0
  CC [M]  /home/abhijeet/OS-Jackfruit/boilerplate/monitor.o
  MODPOST /home/abhijeet/OS-Jackfruit/boilerplate/Module.symvers
  CC [M]  /home/abhijeet/OS-Jackfruit/boilerplate/monitor.mod.o
  LD [M]  /home/abhijeet/OS-Jackfruit/boilerplate/monitor.ko
  BTF [M] /home/abhijeet/OS-Jackfruit/boilerplate/monitor.ko
Skipping BTF generation for /home/abhijeet/OS-Jackfruit/boilerplate/monitor.ko due to unavailability of vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-101-generic'
gcc engine.c -o engine
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$
```

2. Module loaded

```
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo rmmod monitor
sudo insmod monitor.ko
ls /dev/jackfruit_monitor
[sudo] password for abhijeet:
rmmod: ERROR: Module monitor is not currently loaded
/dev/jackfruit_monitor
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$
```

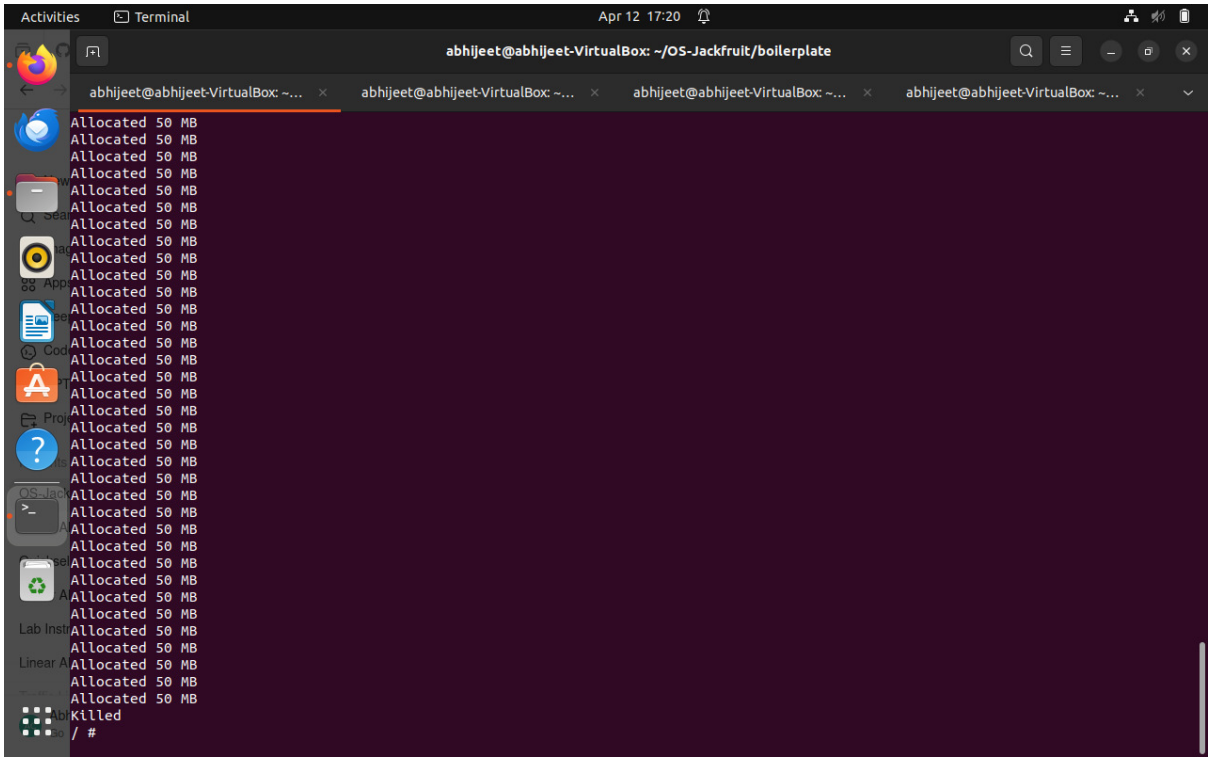
3. memory_hog compiled

```
[sudo] password for abhijeet:
rmmod: ERROR: Module monitor is not currently loaded
/dev/jackfruit_monitor
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$ gcc -static memory_hog.c -o memory_hog
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$ cp memory_hog ../rootfs-alpha/
chmod +x ../rootfs-alpha/memory_hog
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$
```

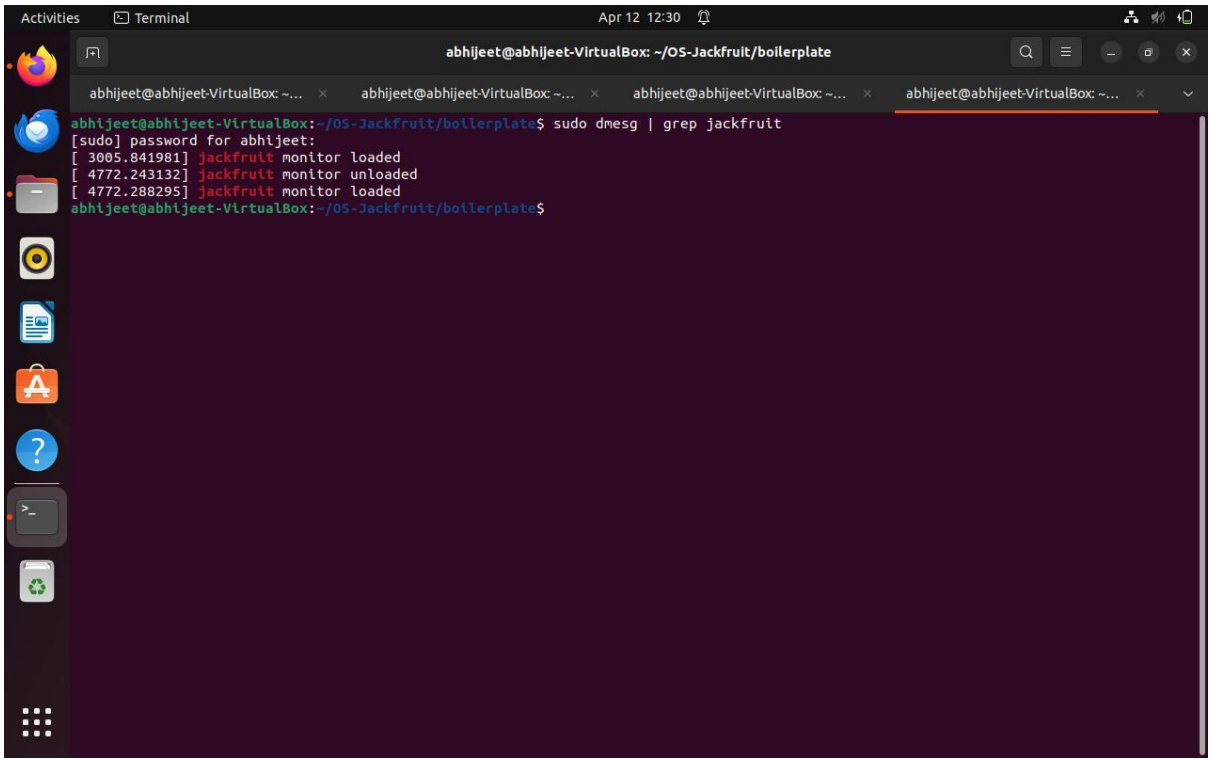
4. Container started

```
abhijeet@abhijeet-VirtualBox: ~        abhijeet@abhijeet-VirtualBox: ~/OS-Jac...
abhijeet@abhijeet-VirtualBox:~$ cd ~/OS-Jackfruit/boilerplate
sudo ./engine ../rootfs-alpha
[sudo] password for abhijeet:
/ #
```

5. Hard-limit kill



6. Kernel logging



7. Multi container

```
abhijeet@abhijeet-VirtualBox: ~        abhijeet@abhijeet-VirtualBox: ~/OS-Jac...
abhijeet@abhijeet-VirtualBox:~$ cd ~/OS-Jackfruit/boilerplate
sudo ./engine ../rootfs-alpha
[sudo] password for abhijeet:
/ #
```

```
abhijeet@abhijeet-VirtualBox: ~        abhijeet@abhijeet-VirtualBox: ~/OS-Jac...
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$ cd ~/OS-Jackfruit/boile
rplate
sudo ./engine ../rootfs-beta
[sudo] password for abhijeet:
/ #
```

8. Metadata tracking

```
/ # exit
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$ ps aux | grep engine
abhijeet    1431  0.0  0.0 163620  3832 ?        Sl   11:04   0:03 /usr/libexec/
ibus-engine-simple
abhijeet   11454  0.0  0.0   9212  2596 pts/0    S+   19:35   0:00 grep --color=
auto engine
abhijeet@abhijeet-VirtualBox:~/OS-Jackfruit/boilerplate$
```

ebra Project Ideas

Screenshot showing no running containers